

Oseme Ochei

Senior Software & Machine Learning Engineer

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Summary

Senior Software & Machine Learning Engineer and AI Researcher with a decade of experience spanning deep learning, backend software development, systems programming, and computational neuroscience. I design, optimize, and deploy production-ready AI systems while also building the surrounding software infrastructure - including APIs, distributed services, and high-performance backend components. Focused on bridging ML research with real-world engineering, I deliver solutions that are testable, scalable, and maintainable. Skilled in Python, Rust, Go, C#, Java, and Node, I integrate ML models into robust software ecosystems, ensuring reliability, reproducibility, and performance at scale.

Tech Skills

Programming & Systems: Python, Rust, C++, Go, Java, C#, JavaScript/TypeScript, SQL, R, MATLAB, Shell scripting

AI / ML & Tooling: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Keras, OpenCV, OpenAI, FastAI, JAX/Flax, Hugging Face, LangChain, LlamaIndex; train-from-scratch & fine-tune pipelines; unit & integration test design; OpenAI APIs

Data & Infrastructure: SingleStore, PostgreSQL, MySQL, MongoDB, FAISS, Pinecone; Databricks, Spark; version-controlled data systems

Web: Django, Flask, FastAPI, Node.js, NestJS, RESTful APIs, GraphQL, .NET, Spring Boot, PHP/Laravel, React, Web Scrapping

Cloud & DevOps: AWS (SageMaker, Lambda, EC2), GCP (Vertex AI, BigQuery, Cloud Functions), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI), Jenkins, Agile, Azure

Testing & Reliability: Cypress, Selenium, Junit, NUnit, PyTest, Jest, Postman, SuperTest, Test-driven ML development, data drift detection, performance benchmarking, automated evaluation pipelines, reproducibility protocols

Specialized Areas: LLMs, computer vision, bioinformatics, production systems at scale, MLOps, experiment design & A/B testing, marketing analytics, client collaboration, documentation

Work History

Senior Machine Learning Engineer | Mathpix | Brooklyn, NY | 03/2025 – Present

- Designed, trained, and deployed advanced computer vision and OCR models for the recognition of complex chemical diagrams, enabling automatic parsing of molecular structures with a 27% accuracy improvement over prior baselines.
- Converted legacy PyTorch segmentation networks to JAX/Flax for TPU execution, reducing training cycle time by 35% and lowering compute cost while maintaining parity in accuracy metrics.
- Implemented large-batch inference pipelines for the OCR service, achieving 4× throughput gains and supporting millions of daily document predictions across international user bases.
- Discovered and resolved a critical ReLU malfunction in fused layer blocks, tracing the issue to an incorrect graph optimization routine and reengineering the affected modules for numerical stability.
- Collaborated directly with high-profile clients such as Google DeepMind, providing tailored model architectures and deployment guidance for their specialized GPU/TPU clusters.
- Developed continuous testing and regression validation pipelines that automatically benchmarked every model revision, ensuring zero degradation in production precision and recall.
- Introduced a hybrid model-management workflow using Dockerized environments and cloud orchestration, improving reproducibility and scalability of the deployment ecosystem.
- Led internal workshops on JAX optimization and vectorized model parallelism to upskill engineering teams.
- Tech: Python, JAX, Flax, PyTorch, GCP, Docker, TPUs, Deep Learning, OCR, Computer Vision, Rust

Senior Data Scientist | IDEXX Laboratories | Dallas, TX | 09/2024 – 02/2025

- Directed the end-to-end design, training, and deployment of embedded computer-vision algorithms for next-generation veterinary diagnostic instruments, achieving over 94% detection accuracy.
- Enhanced Databricks-based pipelines for data ingestion, feature extraction, and model evaluation, cutting overall experimentation time by 40%.
- Built real-time inference modules optimized for NVIDIA GPUs, reducing model latency by 18% while maintaining hardware efficiency across diverse devices.
- Planned a data-driven approach to monitor nucleated cell counts using computer vision techniques, enabling

streamlined quantitative diagnostics for veterinary samples.

- Established cross-validation frameworks and blind test procedures to ensure generalization of deployed models, aligning accuracy metrics with internal clinical validation protocols.
- Worked closely with biomedical engineers and data analysts to translate statistical outputs into meaningful clinical indicators used in product decision-making.
- Utilized robust testing and QA frameworks that achieved a 99.9% regression-free deployment success rate across iterative model updates.
- Authored comprehensive documentation on model versioning, evaluation procedures, and compliance guidelines for audit and regulatory review.
- Tech: Python, Databricks, Spark, Pandas, OpenCV, ML Ops, Embedded Vision, Model Monitoring

Software & ML Engineer | The Bracket Studio | Remote | 07/2020 – Present

- Conducted LLM training, fine-tuning, and evaluation on large datasets across platforms such as Remotasks, Outlier, and Alignerr, contributing to the improved reasoning performance of multiple commercial LLMs, integrated them into scalable backend services supporting millions of daily users.
- Engineered an end-to-end machine-learning pipeline for shelf monitoring in Home Depot retail environments using YOLO and GCP services, achieving a 40% improvement in shelf-stock detection speed.
- Built a GPU-accelerated adaptation of an existing CPU-bound computer-vision library, realizing 6.5× runtime acceleration for large-scale batch inference.
- Partnered with Amazon Bedrock to fine-tune proprietary large-language models for BEN AI, designing structured knowledge indexes via LlamaIndex and SingleStore that enhanced query comprehension and response coherence by 30%.
- Managed formal experimentation and benchmarking for The Fund for Alignment Research, using Weights & Biases to log hyperparameter sweeps and visualize cross-model alignment outcomes.
- Crafted a semi-simplified proof assistant that verified correctness of program logic using symbolic reasoning frameworks, contributing to ongoing ML-for-verification research.
- Produced an adaptive testing engine for nurse training that put into practice reinforcement learning to recommend next questions dynamically, improving exam pass rate by 25%.
- Wrote and curated question sets for TripleByte and Snorkel AI, assessing general coding logic and constructing prompt datasets for LLM robustness testing.
- Mentored junior developers on production-grade React/Node/Python development practices and guided bootcamps on machine-learning fundamentals and data-driven engineering.
- Maintained robust testing infrastructure with CI/CD integration using Docker and Kubernetes, automating QA for ML inference services.
- Tech: Python, PyTorch, LlamaIndex, SingleStore, GCP, Docker, Kubernetes, NLP, Computer Vision, MLOps, Java/Spring Boot, Backend, C#/.NET, JavaScript, AWS, Node, Azure, Agile, AWS, Rust, C++, Go

Senior Research Engineer | Modulate AI | Cambridge, MA | 08/2022 – 08/2023

- Owned the development of Modulate's toxicity-scoring system for voice-chat moderation, integrating multi-modal signals (text, emotion, demographic metadata) to classify conversational harm with 99%.
- Enhanced detection precision by 20 + % through data re-balancing, feature weighting, and probabilistic calibration, enabling more accurate escalation decisions in real-time moderation.
- Devised robust cross-validation and A/B testing frameworks to benchmark model changes before deployment, ensuring parity between experimental and production performance.
- Constructed internal evaluation tools that monitored model drift and optimized regression testing for every release candidate.
- Boosted model explainability by introducing interpretable intermediate metrics that linked emotion scores and textual toxicity classifications.
- Worked closely with engineering teams to integrate streaming inference APIs and to optimize response time within divided systems.
- Contributed to dataset annotation design to reflect reclaimed language and context-specific vernacular in multi-speaker environments.
- Presented model audit reports and research findings to leadership, supporting data-driven product roadmaps.
- Tech: Python, NumPy, Pandas, Rust, NLP, NoSQL/SQL, GitLab, Model Evaluation, A/B Testing

Software Engineer | DoltHub | Dallas, TX | 12/2020 – 12/2021

- Managed DoltHub's data bounty initiatives, where community contributors built structured public datasets using Dolt's version-controlled SQL database.
- Shaped competition schemas, SQL architectures, and bounty workflows that streamlined participant contributions and data merging, improving participation efficiency by 35%.
- Composed Python and Go utilities for high-volume data validation, enabling real-time integrity checks on data merges exceeding 100 GB.
- Deployed testing routines to verify merge consistency and foreign-key integrity across dispersed repositories.
- Teamed up with community participants on Discord to resolve submission conflicts, provide schema feedback, and review pull requests for adherence to standards.

- Built visualization dashboards tracking progress, submissions, and leaderboard standings, improving engagement metrics by 2x.
- Partnered with marketing and product teams to increase visibility of bounty contests and drive user growth across open-source channels.
- Tech: Python, Go, SQL, Pandas, MySQL, Git, Shell Scripting, Rust

Bioinformatician & Software Engineer | SelfDecode | Dallas, TX | 12/2019 – 12/2020

- Oversaw large-scale genomic data analysis, processing multi-terabyte datasets to compute polygenic and behavioral risk scores for complex diseases such as cancer, cardiovascular disease, and COPD.
- Structured pipelines in Python and R to perform variant aggregation, normalization, and statistical inference for health risk modeling.
- Refined conversion methods that translated complex quantitative scores into human-interpretable insights, improving user comprehension and engagement.
- Established a COVID-19 risk assessment tool that modeled symptom severity probability based on epidemiological and biometric factors.
- Validated predictive models using cross-validation and clinical literature correlation, ensuring medical plausibility of statistical results.
- Reduced manual steps of pipeline components for genotype parsing and report generation, reducing manual computation time by 65%.
- Produced detailed documentation and scientific summaries for internal stakeholders and publication review boards.
- Tech: Python, R, C++, NumPy, Pandas, Statistics, Decision Trees, Bioinformatics

Software Engineer | Google | Mountain View, CA | 06/2014 – 07/2015

- Contributed to the Third-Party Actions team, enabling external developers to integrate with Google Assistant via intent binding and query routing frameworks.
- Devised and executed the Nest Thermostat integration, connecting IoT device APIs to Google's smart-home ecosystem.
- Performed extensive grammar engineering and intent classification, improving NLP routing accuracy for user queries by 30%.
- Maintained production services within one of the world's largest codebases, implementing safe rollout and regression testing strategies.
- Supported across product and UX teams to refine natural-language understanding models for contextual command handling.
- Participated in architecture reviews, ensuring system scalability, fault tolerance, and adherence to Google coding standards.
- Drafted internal documentation and SDK references that guided external partner integrations.
- Tech: Python, Java, C#, C++, JavaScript, HTML, NLP, MySQL, Protocol Buffers

Soft Skills

Structured thinking · Leadership · Creative solution design · Strategic Planning · Collaboration · Communication · Fast Learner · Initiative · Mentorship · Continuous Learning

Education

University of Texas at Dallas | *Ph.D. Candidate in Computer Science* | Richardson, Texas | 2018 – 2019

University of North Texas | *Master's Degree in Artificial Intelligence* | Denton, Texas | 2023 - Present

University of Texas at Dallas | *Master's Degree in Systems Engineering and Management* | Richardson, Texas | 2017 – 2018

Baylor College of Medicine | *Doctor of Medicine* | Houston, Texas | 2015 - 2017

Duke University | *Bachelor's Degree in Computer Science* | Durham, North Carolina | 2009 - 2013

- B.S. in Neuroscience
- Minor in Music
- Minor in Japanese

Certifications

Creating a Captive Sphere | *LinkedIn* | 01/2022

C++: Advanced Topics | *LinkedIn* | 12/2021

Apache Spark Essential Training | *LinkedIn* | 07/2021

Docker for Developers | *LinkedIn* | 07/2021

HTML Essential Training | *LinkedIn* | 07/2021

Applied Machine Learning: Foundations | *LinkedIn* | 06/2021

MySQL Advanced Topics | *LinkedIn* | 01/2021

Advanced NLP with Python for Machine Learning | *LinkedIn* | 12/2020
MySQL Essential Training | *LinkedIn* | 12/2020
Triplebyte Certified Generalist Software Engineer | *Triplebyte* | 03/2019

Research

Nicolelis Brain/Machine Interface Labs | Durham, NC | 2011 – 2013

The Nicolelis Lab is known for pioneering work in neuronal population coding, Brain-Machine Interfaces (BMI), and neuro-prosthetics in humans and non-human primates.

- Planned and implemented computational models to prototype neuro-prosthetic systems, improving decoding accuracy of motor signals by ~15%.
- Fine-tuned linear regression and Kalman Filter models to predict motor intentions of rhesus macaques from neuronal activity with more than 85% prediction accuracy.
- Programmed integration for a miniature mind-controlled car, validating BMI signal control in real-time experiments.
- Composed performance data across 100+ trial sessions, deriving quantitative measures of BMI channel capacity for thesis research.
- Authored and exhibited senior thesis on information-theoretic modeling of artificial tactile stimulation, contributing to peer-reviewed research within the lab.

Johns Hopkins Center for Language and Speech Processing | Baltimore, MD | 2012 - 2012

A 6-week international research workshop advancing speech and language engineering.

- Engineered NLP pipelines for multi-speaker audio summarization, reducing document length by 60% while retaining key semantic content.
- Mapped out algorithms to reduce Graph Edit Distance to Binary Linear Programming, improving computational efficiency by ~25% in experimental tasks.
- Calculated and visualized Kullback-Leibler and Jensen-Shannon divergence for more than 10,000 parse trees, enabling comparative syntactic analysis.
- Co-worked with a 10-member international research team to prototype novel language models.
- Displayed results to sponsors including NSF, ODNI, and Google Research, contributing to post-workshop publications and tools used in ongoing NLP research.